

Smart IoT Automation System (Real-World Mini Project)

Embedded Systems and IOT Task-3

Project Report on

SMART LIGHTING USING LDR WITH LCD DISPLAY

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Project Domain:

Embedded Systems & Internet of Things (IoT)

Smart Lighting System Using LDR With LCD Display

The Smart Lighting System using LDR with lcd display is an advancement to the smart lighting system (previous task project). It is an automated lighting solution designed to control a light source based on ambient light intensity. The system uses an Arduino microcontroller to monitor light intensity through an LDR sensor and display the values and state of the light bulb. It provides the user to know about the state of light and intensity of light using an LCD display integrated with i2c.

Components Required

Component Name	Quantity	Purpose
Arduino Uno	1	Main microcontroller used to process sensor data and control the LED.
Light Dependent Resistor (LDR)	1	Detects ambient light intensity.
4.7 k Ω Resistor	1	Forms a voltage divider circuit with the LDR.
LED	1	Represents the lighting system output.
220 Ω Resistor	1	Limits current through the LED and protects it from damage.
Lcd display with i2c	1	Used to display the values and state of the system.

Breadboard

1

Used for temporary circuit connections.

Working Principle

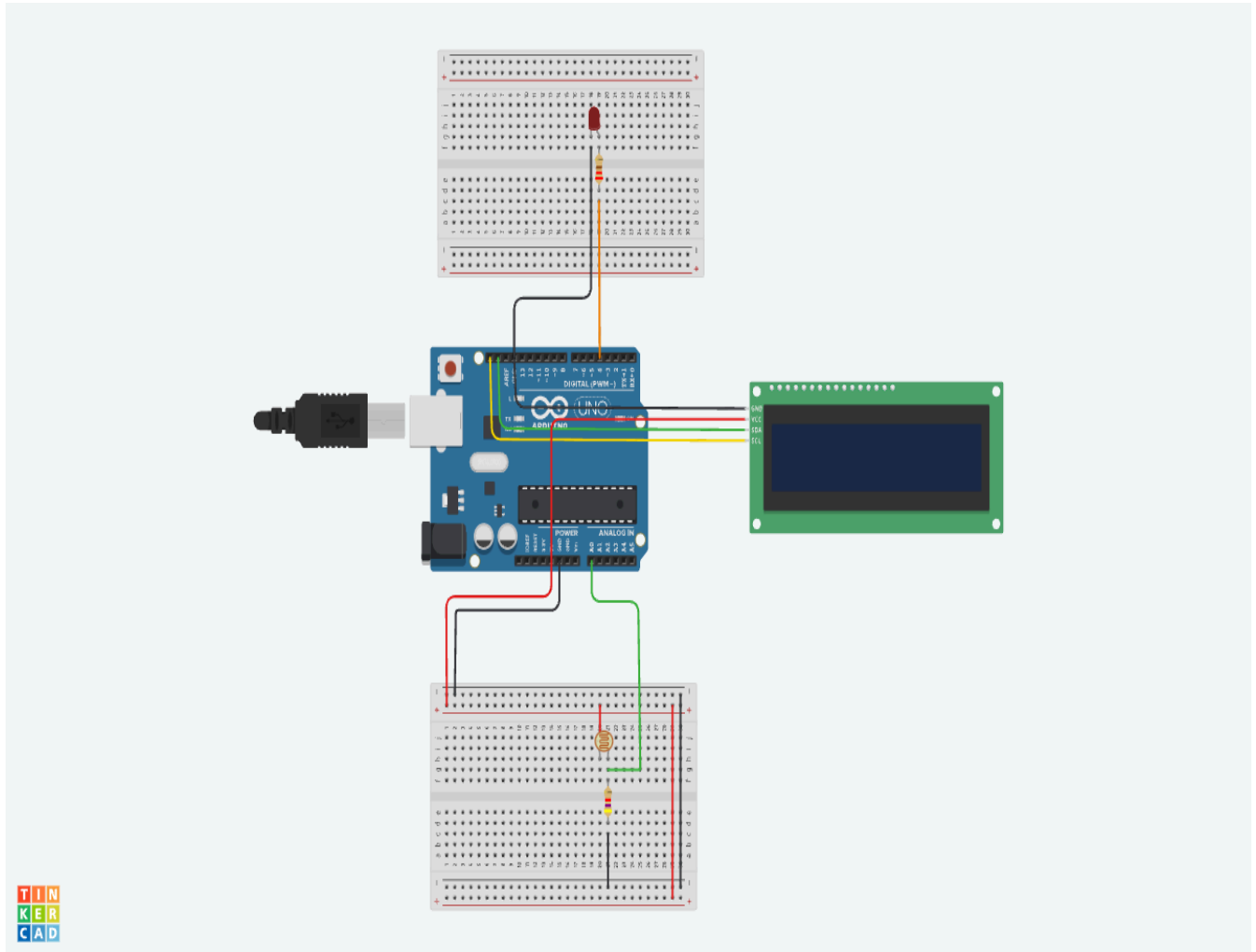
The Smart Lighting System using LDR with LCD display is a real life application in day to day life. It operates based on the intensity of ambient light detected by a Light Dependent Resistor (LDR). The resistance of the LDR varies according to the amount of light falling on its surface. When the surrounding environment is bright, the resistance of the LDR decreases, resulting in a higher voltage. When the environment becomes dark, the resistance increases, resulting in a lower voltage.

The LDR is connected to the Arduino Uno through a voltage divider circuit with 4.7k ohm resistor. The Arduino continuously reads the analog voltage from the LDR using its analog input pin (A0). The obtained value is compared with a threshold value programmed in the Arduino.

If the measured light intensity is below the threshold value, indicating low-light or dark conditions, the Arduino activates the output pin and turns the LED on connected at pin 4. When the light intensity rises above the threshold value, indicating sufficient brightness, the Arduino deactivates the output pin and turns the LED off.

This automatic process continuously repeats for every 500 milliseconds, allowing the system to respond to changes in environmental lighting conditions without human intervention. The system helps conserve energy, improves convenience, and provides an efficient solution for automatic lighting control in real life.

Circuit Diagram



Circuit diagram of smart lighting using LDR with LCD display

Arduino IDE Code

```
// smart lighting using LDR with LCD display
```

```
#include <Adafruit_LiquidCrystal.h>
```

```
const int ldr_pin = A0; // LDR sensor connected to Analog pin A0
```

```
const int ledPin = 4; // LED or Relay connected to Digital pin 13
```

```
int threshold = 500; // Light threshold (adjust based on your environment)
```

```
Adafruit_LiquidCrystal lcd_1(0);
```

```
void setup()
```

```
{
```

```
  pinMode(ledPin, OUTPUT);
```

```
  pinMode(ldr_pin, INPUT);
```

```
  Serial.begin(9600); // Start serial monitor to read LDR values
```

```
  lcd_1.begin(16, 2); // Initiating i2c lcd screen
```

```
  lcd_1.setBacklight(1);
```

```
  lcd_1.print("smart lighting");
```

```
}  
  
void loop()  
{  
    int ldrValue = analogRead(ldr_pin); // Read the ADC value from ldr (0 to  
    1023)  
  
    Serial.println(ldrValue); // Print value to Serial Monitor  
  
    // IF-ELSE BLOCK  
  
    // If it is dark (sensor value drops below threshold)  
    if (ldrValue < threshold) {  
        digitalWrite(ledPin, HIGH); // Turn the light ON  
        Serial.println("light on!");  
        lcd_1.setCursor(0, 1);  
        lcd_1.print("LIGHT ON"); // print on lcd  
    }  
  
    // If it is bright  
    else {  
        digitalWrite(ledPin, LOW); // Turn the light OFF  
        lcd_1.setCursor(0, 1);  
        lcd_1.print("LIGHT OFF"); // print on lcd  
    }  
  
    delay(500);  
}
```

Code Explanation

The Adafruit Liquid Crystal header file is included to enable communication with the 16×2 LCD display.

The LDR sensor is connected to analog pin A0 to measure ambient light intensity.

An LED or relay is connected to digital pin 4 for automatic light control.

A threshold value of 500 is defined to distinguish between dark and bright conditions.

In the `setup()` function, the input and output pins are configured, serial communication is started, and the LCD is initialized.

The LCD displays "smart lighting" when the system powers on.

In the `loop()` function, the Arduino continuously reads the LDR value using the `analogRead()` function.

The sensor reading is compared with the threshold value using an if-else condition.

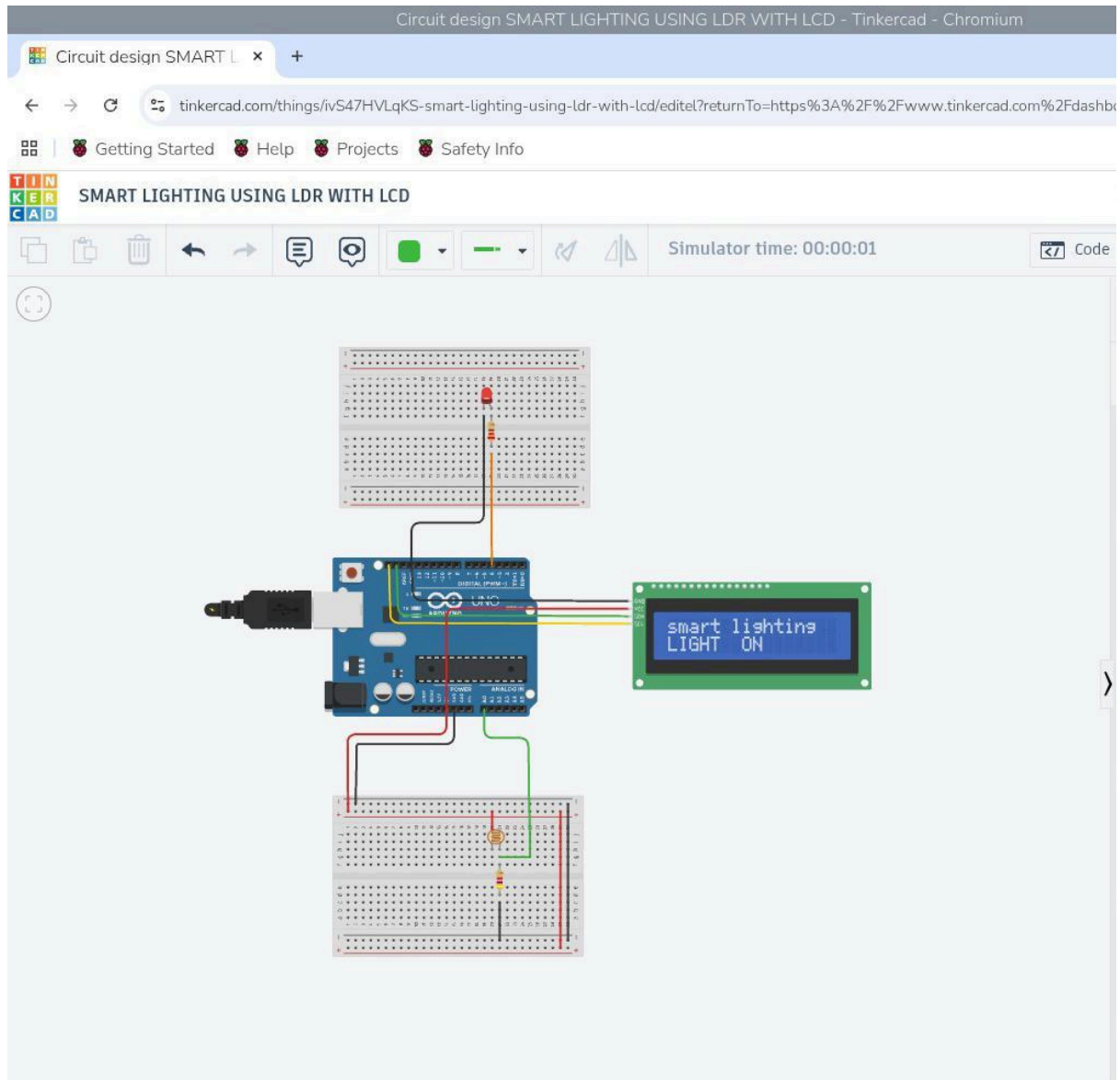
When the LDR value is below 500, the environment is considered dark, and the LED is turned ON. The LCD displays "LIGHT ON".

When the LDR value is above or equal to 500, the environment is considered bright, and the LED is turned OFF. The LCD displays "LIGHT OFF".

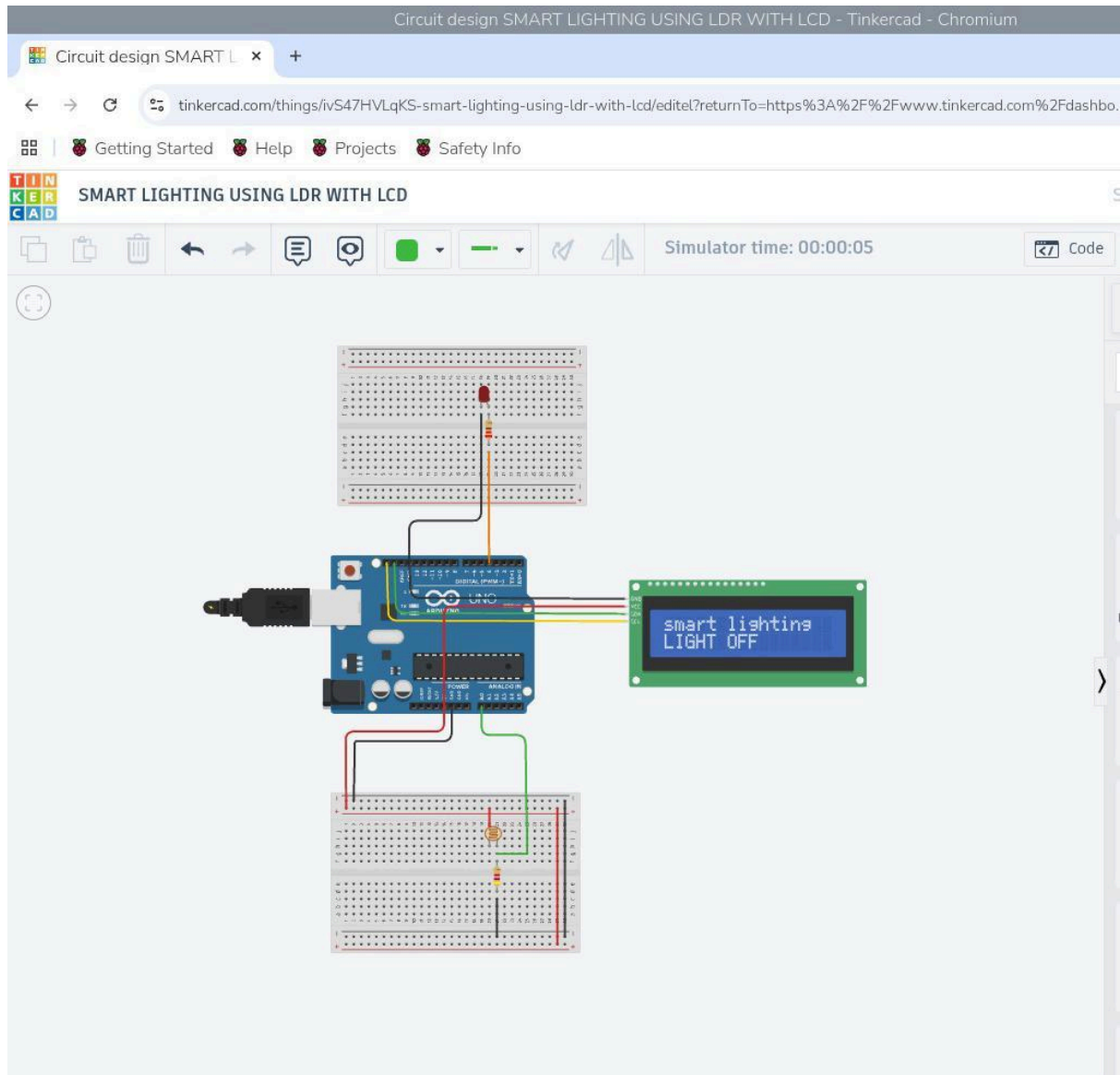
The system continuously monitors light intensity and automatically controls the lighting, helping to save energy and improve convenience.

A delay of 500 ms is provided between successive readings to ensure stable operation.

LCD OUTPUT:



Output of lcd showing LIGHT ON when the brightness of light is higher than threshold value and turns the LED ON.



Output of lcd showing LIGHT OFF when the brightness of light is lower than threshold value and turns the LED OFF.

SERIAL MONITOR OUTPUT:

```
11 void setup()
12 {
13   pinMode(ledPin, OUTPUT);
14   pinMode(ldr_pin, INPUT);
15
16   Serial.begin(9600); // Start serial monitor to read LDR values
17
18   lcd_1.begin(16, 2); // Initiating i2c lcd screen
19   lcd_1.setBacklight(1);
20   lcd_1.print("smart lighting");
21
22 }
23
24 void loop()
25 {
26   int ldrValue = analogRead(ldr_pin); // Read the ADC value from
27   Serial.println(ldrValue); // Print value to Serial Monitor
28
29   // if-else block
30   // If it is dark (sensor value drops below threshold)
31   if (ldrValue < threshold) {
32     digitalWrite(ledPin, HIGH); // Turn the light ON
33     Serial.println("light on!");
34     lcd_1.setCursor(0, 1);
35     lcd_1.print("LIGHT ON"); // print on lcd
36   }
37   // If it is bright
38   else {
39     digitalWrite(ledPin, LOW); // Turn the light OFF
40     lcd_1.setCursor(0, 1);
41   }

```

Serial Monitor

```
473
light on!
473
light on!
535
814
814
...
```

Send Clear

Output of serial monitor printing the values of LDR circuit.